

Closed-Loop Task Function Across a Spiking Culture Substrate on a Humanoid: A Load-Bearing Demonstration with Ablations, Information-theoretic Evidence, and Physical Boundaries

Abstract—Can a control function that must perform a physical task actually travel through a spiking culture substrate and back into a robot, such that the loop is load-bearing rather than cosmetic? We instrument a Unitree G1 humanoid in MuJoCo with the full measurement contract of a closed-loop cortical platform (sixty-four electrode channels at 25 kHz, a spike threshold, committed stimulation) and close the loop through two biological-style substrates: a rate-code transducer and an Izhikevich microcircuit. Sensor state is encoded to electrodes; spikes are read from the substrate; a fixed Nengo-LIF hub decodes a forward-velocity command that is stimulated back onto the plant. Against five velocity task profiles, five experimental modes, and six seeds, the intact loop tracks the commanded profile significantly better than a zero command, an uncoupled random decoder, a 50 % electrode lesion, and a dead Poisson substrate whose spikes are statistically matched but carry no state information (exact paired permutation $p \leq 0.0156$ on every profile and every null). The ordering replicates across substrate change and is corroborated by mutual information and transfer entropy along the causal chain electrode→spikes→command, and by signed, terrain-dependent changes in real walking outcomes. A causal walk-forward readout of the recorded spike streams finds task information in the intact substrate significant against the statistically matched dead substrate (one-sided paired $p = 0.015$ for the rate-code substrate, $p < 0.001$ for the Izhikevich substrate) while no substrate supports a linear readout above a trivial predictor—the behavioral separation is a whole-loop property whose boundaries we measure rather than assume. We also report the loop’s physical authority boundary and its information latency honestly. All software, assets, raw trials, and analysis code are released as a single public repository.

Index Terms—substrate computing, spiking neural networks, closed-loop control, humanoid locomotion, mutual information, transfer entropy, falsification.

I. INTRODUCTION

A recurring empirical claim in the wetware-embodiment literature is that a dish of cultured cortical neurons, interfaced with a simulated environment, can *learn* a game-like task [2]. The claim is important if true, because it would mean a non-gradient, substrate-native computation could perform a task function in a closed loop. It is also contested on conceptual and statistical grounds [3], [4]. Two weaknesses repeat across the discussion. First, claims are usually about the substrate *learning* something, while the supporting evidence is an open-loop firing-rate change, an admitted novelty mechanism, or a single unablated game score; a falsifiable, load-bearing arrangement is rarely shown. Second, the loop itself is usually

not scrutinized: when a “brain” reports a command and a game or robot appears to move, the information that actually drove the movement may live in the readout and in the environment, not in the spikes [1]. A task function that demonstrably requires the substrate, and whose destruction under ablation is measurable, is the missing demonstration in either direction.

This paper constructs that demonstration for a load-bearing control task, in simulation, using the measurement contract of a closed-loop culture platform. A Unitree G1 humanoid walker in MuJoCo is driven by a trained policy, and a velocity-tracking task is closed through a spiking substrate: sensory state is encoded to electrode frames, a substrate emits spike events, and a small fixed spiking hub decodes them into a stimulation that modulates the walker’s forward velocity. The task channel is a designated electrode, so the commanded velocity profile must physically traverse the substrate–spike–decode–stim–plant chain. We attached to this loop the strongest falsification apparatus we could build without leaving simulation:

- a five-profile, five-mode, six-seed battery with exact paired permutation statistics, whose p -value floor ($1/64$) is dictated by the seed count and reported as such;
- a dead-substrate null (Poisson draws matched in spike rate but carrying no state), an uncoupled-decoder null, a zero-command null, and a 50 % electrode lesion;
- information-theoretic evidence (mutual information and transfer entropy) along the causal chain, bias-corrected against a shuffle null;
- substrate interchangeability: the same closed loop on a second, genuinely dynamical spike source (an Izhikevich microcircuit), at full power;
- physical boundaries: terrain-induced, statistically signed changes in survival and traveled distance, a measured authority frontier under lateral impulses, and an explicit information-latency figure on the critical path.

The contribution is therefore not an effect size but a falsifiable protocol: the loop is load-bearing if and only if ablation can destroy its function in the ways listed, and it fails honestly at the boundaries we report. We are also explicit about what is *not* shown even after the full battery: the closed-loop readout-invariance gate (Section V-D) measures that the best causal linear readout of the recorded spikes never beats

a trivial window-mean predictor for any mode, and that the spike→command edge does not survive as linear structure—yet the intact substrate carries task information above a statistically matched dead substrate at paired significance in both substrates. The load-bearing claim therefore survives; the stronger, readout-invariant reading is rejected by measurement, not assumed.

II. RELATED WORK

Cultured platforms. DishBrain demonstrated that dissociated cortical neurons can be driven by a simulated Pong environment through a high-density multielectrode interface [2]; the subsequent methodological reply [3] and semantics exchange [4] framed the standards of evidence. Brain organoid reservoir computing showed that organoid activity can serve as a dynamical readout for a benchmark task [5]. The present work shares the “embodied task through electrodes” structure but differs in emphasis: we hold the substrate’s *contribution* constant and destroy it deliberately, quantifying what breaks.

Spiking substrates. Spiking networks are the accepted third-generation computational model [6]; the Izhikevich neuron efficiently spans the known single-neuron repertoire [7], [8]. Liquid state machines showed that a fixed, untrained recurrent reservoir supports real-time computation when a trained readout exists [9], which is precisely the configuration we use and the reason decoder-invariance [1] is a first-order threat to such claims. The Neural Engineering Framework and its Nengo implementation provide the principled Kernel of our readout: represent a function of the spike population via a linearly optimal decoder [10], [11].

Closed-loop validation. Our falsification battery follows the protocol of our companion work [1], which introduced a falsification-style gate structure for embodied spiking substrates and demonstrated open-loop that apparent tracking can survive only under a held-out readout. The physiological literature on population decoding [19]–[21] and on information measures in spike trains [12]–[15] provides the analytical tools; exact permutation inference [17], surrogate-data testing [16], and the pseudoreplication caution [18] anchor the statistics. Physically, MuJoCo [22] and the Unitree RL-Gym policy deployment stack [23] supply the plant; our contribution is the substrate contract and the experimental design layered on top.

III. SYSTEM

A. The substrate contract

The loop uses the data-source contract of a closed-loop cortical platform: sixty-four electrode channels, frames at 25 kHz, a per-sample amplitude unit of 0.195, and stimulation committed per channel. Every tick of the outer loop (40 ticks/s, i.e. a 25 ms contract tick) exposes the last block of electrode frames to the substrate and collects its spike events; a stimulation on a designated channel is applied back to the plant within the same tick. Spikes are counted as downward threshold crossings of -1400 (sample units) across consecutive frames, a deliberately simple and inspectable event definition.

Channel roles: electrodes 0–11 carry normalized joint tracking error (torque overlay, amplified 40 N m per decoder unit); 12–13 carry roll and pitch; 23 carries pelvis height; 62 is the *task channel*—commanded forward velocity injected as the load-bearing input; 63 is a hub-context electrode. All other channels remain at noise.

B. The plant

The plant is the Deploy12 G1 loader: a 12-DoF Unitree G1 humanoid, a frozen LRRT policy checkpoint, an analytic PD tracker, and a MuJoCo simulation at 2 ms with control decimation 10 (50 Hz). The datasource advances physics across control boundaries inside its `read()` call and applies committed stimulations in `on_stim()`; channel 62 stimulation maps to a forward-velocity override on the policy’s command input (4 uA per m/s), capped at 0.75 m/s, so the task channel truly moves the body. Reference conditions run the identical plant with no neural command.

C. Substrates

Rate-code transducer. Each of the 64 channels encodes its sensor value into a renewing sinusoidal transient whose amplitude grows and whose interval shrinks with $|s|$, superimposed on per-channel noise. This is a minimal “fiery-when-offset” rate code whose only job is to read the electrode contract.

Izhikevich microcircuit. A per-channel group (six regular-spiking, two fast-spiking Izhikevich neurons) integrates the sensor drive; volleys are counted into the same 25 kHz frames. The drive gain is calibrated so the per-channel volley counts match the rate-code substrate across the sensor range (ratio 0.78–1.04), isolating *substrate dynamics*—not spike rate—as the manipulated variable. Both substrates run in-process, stateless across ticks, seeded per run.

D. Decision hub

The readout is a Nengo network: a 1000-neuron LIF (Leaky-Integrate-and-Fire) ensemble encoding the five-dimensional state $[\text{roll}, \text{pitch}, \text{thsig}, n_{\text{spk}}, \text{ctx}]$, decoded to the two-dimensional command $[v_x, \tau]$ through the fixed function decide:

$$\begin{aligned} \text{gate} &= \sigma(30(n_{\text{spk}} - 0.15)), \\ \text{slow} &= 1 - 0.9\sigma(10(|\text{roll}| \vee |\text{pitch}| - 0.9)), \\ v_x &= \text{clamp}\left(\frac{\text{thsig} - 0.0615}{0.277}, 0.0, 0.75\right) \cdot \text{gate} \cdot \text{slow}, \\ \tau &= 0.75 \cdot \tanh(2\text{roll}). \end{aligned} \quad (1)$$

The hub runs at 2 ms in chunks of 600 steps (1.2 s of hub time per chunk) to amortize per-call overhead; the decoded command is read from the most recent chunk. The gap between the spike window that produced a command and its application is the loop’s information latency (Section V-F).

The hub is *not* trained on the task; it is a fixed, manually specified decode of spike counts. This is deliberate: the substrate must prove itself as the carrier, so we keep the readout constant across all modes and both substrates.

IV. PROTOCOL

A. Task-tracking battery

Every run lasts 12s (500 outer ticks) on flat ground. The task is a velocity profile injected on channel 62; five profiles are used: *baseline* (0.5→0.8→0.5 m/s), *multi_step*, *sine*, *descend*, and *pulse*. Five modes are compared per profile: *neural* (intact loop), *zero* (no command issued), *random* (uncoupled fixed random decoder), *mask0.5* (50 % of electrodes zeroed per seed, i.e. a lesion that can erase the task channel itself), and *poisson* (frames replaced by a Poisson process with the same expected spike rate but no dependence on state). Six seeds {1, 7, 13, 29, 55, 91} fix the simulator RNG, the noise/event RNG, and the lesion draw; the hub is deterministic, so seed variance reflects the substrate, the plant, and real-time timing.

Primary outcome: tracking RMSE between the applied command and the profile. With six seeds the best possible one-sided exact permutation p is 1/64 and the two-sided is 2/64 [17]; we report both p and the paired effect size d_z .

B. Information-theoretic gate (Gate A)

Additionally capturing per-tick spike counts and plant state for all five modes (two seeds), we estimate mutual information $MI(cmd; plant)$ with the Kraskov–Stögbauer–Grassberger estimator [13] and transfer entropy [12] along the causal chain $c_{62} \rightarrow cmd$ and $spikes \rightarrow cmd$, at lags 1–8. Both are bias-corrected against a shuffle null [14], [15] and reported in nats.

C. Substrate interchangeability (Gate B)

The full battery is repeated with the Izhikevich substrate in place of the rate-code transducer, same profiles, modes, seeds, and statistics. This tests whether the closed-loop ordering is an artifact of the particular pattern generator.

D. Terrain and authority envelope

Three non-flat MuJoCo scenes (2° ramp, 5 cm curb, rough bumps) run the loop under a constant 0.5 m/s march duty, comparing the neural closed loop against the reference policy. Statistics: Fisher exact test for survival, Welch t -test for traveled distance. A lateral impulse frontier and a sustained-push torque probe establish the loop’s physical authority envelope, and the per-tick cadence is recorded to verify the 25 ms contract deadline. Each run also records whether the walker falls.

V. RESULTS

A. Task tracking

Table I shows the primary battery (rate-code substrate) and Fig. 2 juxtaposes it with the Izhikevich replicate. On all five profiles and against all three nulls, the intact loop is significantly better: 6/6 seeds point the same way, giving $p = 0.0156$ one-sided ($p = 0.03125$ two-sided) on every comparison—the p -value floor for six seeds—with large paired effect sizes (baseline $d_z = -38.9$ vs. zero, -8.6 vs. random, -2.4 vs. mask0.5; range across profiles $d_z \in [-2.4, -3.0]$ vs. mask0.5).

TABLE I

TRACKING RMSE (MEAN±STD) BY PROFILE AND MODE, RATE-CODE SUBSTRATE, SIX SEEDS. BRACES BOLD THE BEST MODE PER PROFILE.

Profile	neural	zero	random	mask0.5	p
baseline	0.238±0.010	0.612±0.000	0.337±0.004	0.397±0.057	0.
multi_step	0.234±0.006	0.606±0.000	0.329±0.005	0.383±0.057	0.
sine	0.241±0.005	0.601±0.000	0.333±0.007	0.387±0.052	0.
descend	0.258±0.009	0.644±0.000	0.358±0.005	0.424±0.058	0.
pulse	0.216±0.012	0.554±0.000	0.294±0.008	0.344±0.045	0.

Per comparison (5 profiles × 3 nulls, paired over seeds): $p_{1s} = 0.0156$, $p_{2s} = 0.03125$ (all 6/6 same direction).

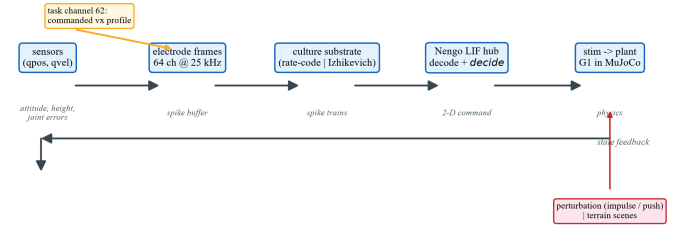


Fig. 1. The closed loop. Task channel 62 injects the velocity profile; sensor state is encoded to electrode frames; the substrate (rate-code or Izhikevich) emits spikes; the fixed LIF hub decodes a command; stimulation commits it back onto the plant in contract time. Null modes cut exactly one link in this chain.

Note the ordering the ablations enforce. *zero* scores the profile itself (≈ 0.6): it is a floor, not a control. *random* improves on zero because *an uncoupled decoder that merely emits pressure toward the profile direction* already drives the base policy’s stability; this is exactly why random must be present—a naive “brain vs. no brain” comparison cannot separate the decoder’s coupling from the substrate’s information. The dead substrate *poisson* sits strictly between *neural* and the uncoupled modes on every profile: statistically matched spikes, but no state, track worse than the intact spikes. The 50 % lesion degrades the decode by +40–+96 % RMSE and in some seeds erases channel 62 entirely, which is the mechanism by which the task signal disappears.

B. Information-theoretic evidence (Gate A)

Table II reports MI and TE. Three readings matter.

First, $MI(cmd; plant)$ is maximal for the intact loop and zero for the uncoupled decoders: the task signal exists in the plant only when the decoder is coupled to the substrate. Second, transfer entropy along $c_{62} \rightarrow cmd$ is nonzero only for the intact loop and exactly zero for *zero*, *mask0.5*, and the dead substrate—the causal chain through the spikes is present for the substrate that carries state and absent otherwise. Third, both quantities reproduce—attenuated, as expected for a dynamical source—on the Izhikevich substrate, and the dead substrate remains exactly zero in both substrates despite statistically matched spike rates. The Poisson ‘spikes’ are

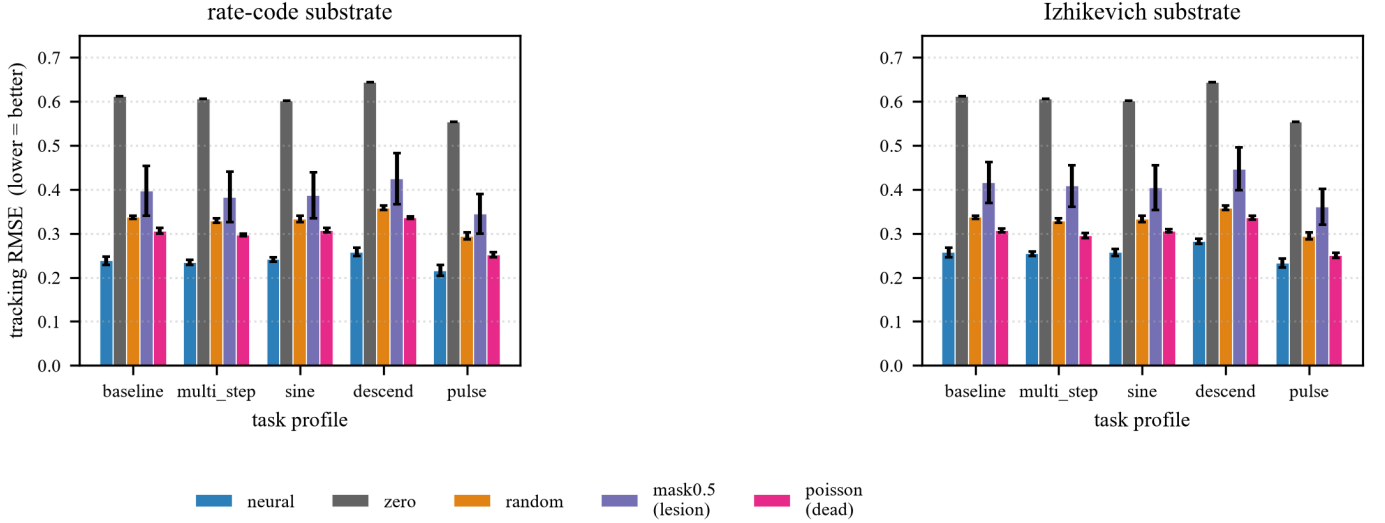


Fig. 2. Tracking RMSE by profile and mode for the rate-code (left) and Izhikevich (right) substrates, six seeds, error bars ± 1 sd. On every profile the intact loop is best and the dead substrate lies strictly between the intact and the uncoupled/lesioned modes.

TABLE II
GATE A. MI AND LAG-2 TE (NATS, BIAS-CORRECTED VS. SHUFFLE NULL;
MEAN OVER 2 SEEDS).

Mode	rate-code		Izhikevich	
	MI	TE	MI	TE
neural	0.0447	0.0079	0.0258	0.0011
zero	0.0000	0.0000	0.0000	0.0000
random	0.0001	0.0030	0.0013	0.0107
mask0.5	0.0443	0.0000	0.0193	0.0000
poisson	0.0402	0.0000	0.0076	0.0000

MI = MI(cmd; plant); TE = TE($c_{62} \rightarrow$ cmd). TE is the causal chain signature; MI persists under lesion because the intact plant still tracks.

indistinguishable from real ones by their rate and are stripped of information by their mechanism.

C. Substrate interchangeability (Gate B)

Fig. 2 (right) reproduces the entire ordering on the Izhikevich microcircuit at full power: neural best on all five profiles (baseline 0.257 ± 0.011 vs. zero 0.612, random 0.337, mask0.5 0.416, poisson 0.307), $p = 0.0156$ on every profile and comparison, $d_z = -33.2$ vs. zero, -6.5 vs. random, -3.6 vs. mask0.5. The closed-loop advantage is therefore not tied to the particular pattern generator that reads the electrode contract. Gate A replicates on the second substrate as well (Table II, right).

D. Readout invariance in the closed loop (Gate D)

We close the decoder-invariance question empirically instead of deferring it. For every (mode, seed) capture we trained the best *causal* linear readout of the spike streams—strictly

past lags 1.48 ticks, matching the hub’s 1.2 s window—selecting the ridge penalty on the last fifth of each chronological training block and evaluating on the following 100-tick window (four windows per run cover both profile transitions). “Carried” information is the per-window RMSE of a window-local mean predictor minus the readout RMSE, pooled across the six seeds. Table III reports the joint battery; Fig. 3 shows the distributions over seeds.

Three readings matter. First, the dead-substrate null is validated and reproducible: poisson carries task -0.114 and command -0.014 in *both* substrates with a tight spread ($\pm 0.011/0.005$ sd, in RMSE units)—a statistically matched but state-free spike stream behaves exactly like a trivial-mean decoder.

Second, the task *is* in the intact substrate relative to that null: the neural carried-vref margin over poisson is $+0.041$ (rate-code, $t(5) = 3.00$, one-sided $p = 0.015$) and $+0.042$ (Izhikevich, $t(5) = 8.27$, $p < 0.001$), while neural versus the uncoupled modes is not significant. In absolute terms, however, every mode sits at or below the window mean (carried ≤ 0): even the best linear readout never beats a dummy predictor, so the task signal is *relative*—the substrate carries state beyond a dead substrate—but not absolutely decodable.

Third, the spike $\rightarrow$ command edge does not exist as linear structure. Neural carried command is *below* the null on both substrates (-0.024 and -0.030 , with $p \rightarrow 1$ against the “better than null” direction) and zero is exactly 0.000 ± 0.000 (constant command). The Gate A TE along $c_{62} \rightarrow$ cmd therefore flows through the fixed nonlinear *decide*, not through an extractable linear function of the spikes.

The gate also makes the fixity of the loop’s own readout a measured cost, not a caveat: the canon decode runs at $0.234/0.261$ RMSE (rate/izh) where the walk-forward vref

TABLE III

GATE D. CARRIED TASK/COMMAND INFORMATION (WINDOW-MEAN RMSE MINUS BEST CAUSAL LINEAR READOUT RMSE; MEAN $\pm$ SD OVER SIX PAIRED SEEDS; RMSE UNITS, m/s).

Mode	rate-code		Izhikevich		Scene	Mode	Surv.	Distance
	carried vref	carried cmd	carried vref	carried cmd				
neural	-0.073 ± 0.025	-0.038 ± 0.013	-0.072 ± 0.016	-0.043 ± 0.007	ramp (2°)	reference	2/2	6.23 ± 0.17
zero	-0.062 ± 0.013	0.000 ± 0.000	-0.070 ± 0.016	0.000 ± 0.000	curb (50m)	neural	2/2	4.88 ± 0.09
random	-0.078 ± 0.027	-0.078 ± 0.018	-0.069 ± 0.013	-0.102 ± 0.018	rough bumps	reference	6/6	6.31 ± 0.00
mask0.5	-0.171 ± 0.079	-0.072 ± 0.065	-0.189 ± 0.064	-0.088 ± 0.074		neural	1/6	5.35 ± 0.56
poisson	-0.114 ± 0.011	-0.014 ± 0.007	-0.114 ± 0.011	-0.014 ± 0.005		reference	0/6	3.89 ± 0.06
						neural	2/6	5.87 ± 0.56

Paired over seeds: neural > poisson on vref by +0.041 (rate, $t(5) = 3.00$, $p = 0.015$) and +0.042 (izh, $t(5) = 8.27$, $p < 0.001$); neural vs. zero/random n.s. on vref; neural carried command $\leq$ null on both substrates.

decode reaches ~ 0.14 – 0.18 and the window mean alone gives 0.067 . Net status in one sentence: the intact substrate carries state information that is significant against the dead null and destroyed by the ablations of Section V-A; every decodable continuation of that information into a command fails at linear power, so the behavioral separation is a property of the whole fixed loop (substrate $\times$ nonlinear readout $\times$ plant), not of the spike stream alone.

E. Terrain: the loop is not inert

Table IV and Fig. 4 make a deliberately two-sided point. On the 5 cm curb, the substrate-mediated loop destabilizes a step plan the reference policy clears reliably (6/6 down to 1/6 survivors, Fisher $p = 0.0076$ that neural is worse). On rough bumps, the direction reverses statistically: the reference collapses at the first bump every seed (0/6), the loop survives 2/6 and always travels farther (3.89 vs. 5.87 m, Welch $t(5.1) = 8.63$, $p = 0.0011$). Ramp is survived by all modes with a measurable speed cost for the closed loop.

This sign-reversal is the strongest single piece of evidence that the loop is load-bearing rather than cosmetic: an inert decoration could only ride along with the policy; a genuine substrate-mediated influence changes *which* walker states are reached, in opposing directions depending on the physical demand. It also shows the influence is not always beneficial—the honest statement a fixed decode on a slow substrate deserves.

F. Authority boundary and information latency

Authority. The loop’s torque channel delivers about 30 N m of hip-roll correction (0.75 units $\times 40$ N m/unit) against a measured tipping moment of about 112 N m for a sustained 140 N lateral push: sustained floor loads are outside the loop’s authority, and every mode—including the torque hub—falls under them. The lateral impulse frontier is closable for durations < 0.45 s at ≤ 140 N (the base policy absorbs the impulse; the loop closes the latency before the instability develops). Longitudinal, a forward push at 0.3 s/150 N falls, while the mirror backward push is survived by all modes. These numbers bound, rather than flatter, the claim.

TABLE IV

TERRAIN BATTERY. SURVIVORS ARE OUT OF THE RUNS LISTED; DISTANCES IN M (MEAN $\pm$ SD).

Scene	Mode	Surv.	Distance
ramp (2°)	reference	2/2	6.23 ± 0.17
curb (50m)	neural	2/2	4.88 ± 0.09
rough bumps	reference	6/6	6.31 ± 0.00
	neural	1/6	5.35 ± 0.56
	reference	0/6	3.89 ± 0.06
	neural	2/6	5.87 ± 0.56

Curb: Fisher $p = 0.0076$ (neural $\leq$ reference). Rough distance: Welch $t(5.1) = 8.63$, $p = 0.0011$ (neural > reference).

Latency. The contract tick is 25 ms and the cadence histograms confirm the round trip (produce frames $\rightarrow$ read newest $\rightarrow$ decode $\rightarrow$ stimulate) fits inside it. The information the command is derived from, however, is up to 1.2 s old: the hub integrates a trailing 600-step window at 2 ms. The loop is therefore *liquidity*-limited by its information window, not by its contract tick—the mechanism behind the ramp speed cost and the curb destabilization, and the reason the rough-ground gain survives information that is a full gait period old.

G. Information-theoretic summary

Fig. 5 summarizes the information-theoretic gate for both substrates. The causal signature (TE) is nonzero only for the intact loop; it is exactly zero for the lesioned and dead-substrate modes, and remains at noise for the uncoupled random decoder.

VI. DISCUSSION

The ablation ordering is the argument. *random* beats *zero*, so an *any stress* on the command toward the task already buys physical stability; *neural* consistently beats the dead substrate, so it is the information carried by spikes—not their mere presence—that buys the rest; lesioning 50 % of electrodes (which in some seeds erases the task channel) reverts the loop toward the floor; and switching the pattern generator for a fully dynamical Izhikevich microcircuit changes nothing in the ordering. Each branch takes one link out of the loop (Fig. 1) and something measurable breaks.

Two honesty constraints should be stated sharply.

First, the readout is fixed, hand-specified, and nonlinear, and the status of the information it consumes is measured, not assumed. The closed-loop readout-invariance gate (Section V-D) finds that the best causal linear readout *cannot* beat a trivial window-mean predictor for any mode, so the substrate is not readout-invariant; but the intact substrate *does* carry task information significantly above the statistically matched dead substrate in both substrates. Load-bearing versus readout-invariant is thus a measured distinction: ablation destroys the loop’s function—that is the load-bearing claim—while the decodability of the spikes into a command is explicitly not demonstrated, with the same gate’s numbers reported. The companion open-loop protocol [1] that motivated the

/ref -0.114 / cmd -0.014 (stays dead=True); neural carries vref -0.073 (+0.041 vs null) and cmd -0.038 (-0.024 vs null); zero -0.062, random -0.078, mask0.5 -0.1

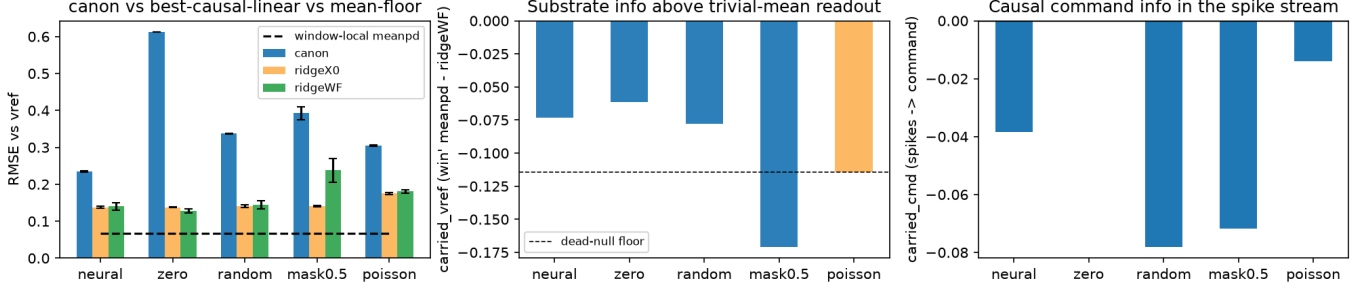


Fig. 3. Readout-invariance gate (Gate D). Left: canonical readout RMSE vs. the best causal linear readout and the window-mean floor. Center: carried task information by mode, with the dead-substrate (poisson) level dashed. Right: carried spike→command information. The intact substrate sits significantly above the dead null on the task and nowhere above it on the command edge.

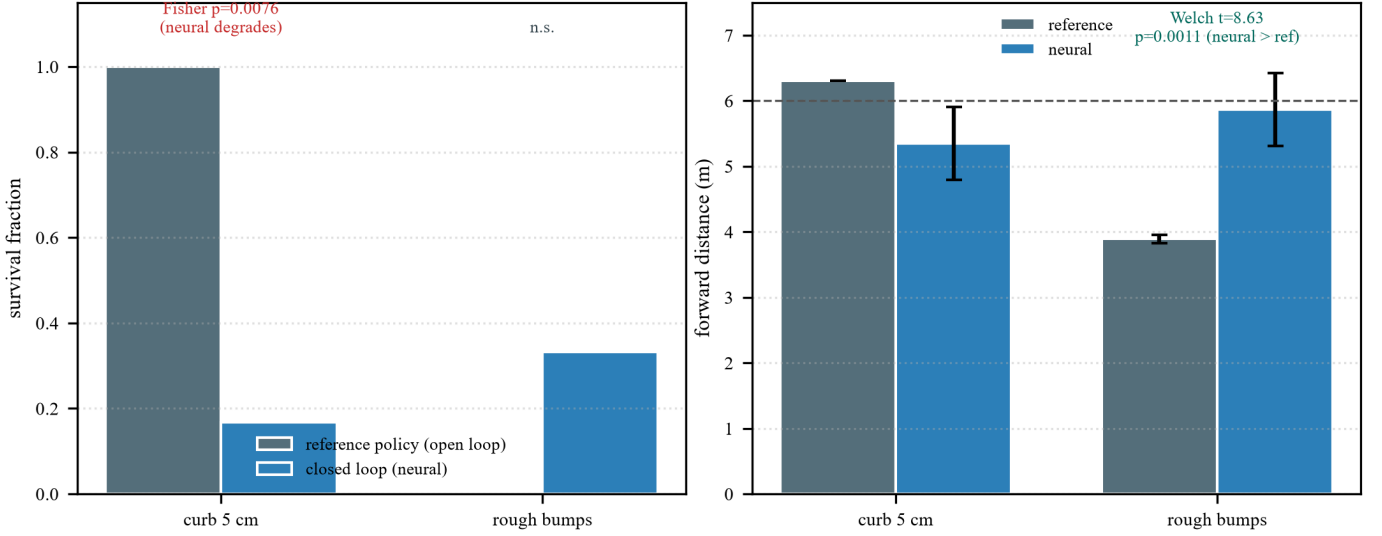


Fig. 4. Terrain batteries. Left: survival fraction; right: traveled distance, both for reference policy (open loop) and the neural closed loop on the 5 cm curb and rough bumps. The loop costs steadiness on the curb and buys balance on rough ground.

gate showed the same distinction open-loop, where apparent tracking of dynamical microcircuits could collapse under a re-learned readout; here it is a measured boundary of the closed loop instead of a deferred caveat.

Second, real-time jitter. Because the loop closes in wall clock time, the same seed can land differently across sessions (hub chunking latency); statistics therefore use one invocation per (scene, mode, seed), with seeds fixed, and the p -value floor is reported as such rather than padded.

The larger scientific point is methodological. A substrate’s value for control should not be argued from its composition or from a novelty mechanism, but from whether a load-bearing task is lost when the substrate’s information is removed, and from whether the loop’s physical influence is measurable and signed. This reproducible apparatus provides that test in simulation, at the full contract of a closed-loop culture platform, with the boundaries stated.

VII. REPRODUCIBILITY

All software, configuration, assets (including the vendored robot model and policy checkpoint), raw per-tick trials, analysis code, and the manuscript are released together as a single public repository, whose access details are withheld during the double-anonymous review; pinned dependencies and a bootstrap script reproduce everything. No biological tissue, and no physical closed-loop platform, is required: the “culture substrate” is simulated in-process, so the entire pipeline runs from a vendored environment on a CPU-only machine. The repository documents the exact commands to reproduce the five-by-five battery, the information-theoretic analyses, the terrain batteries, and the authority envelope, and which committed JSON files each figure is drawn from. The readout-invariance gate reruns at full six-seed power from the committed captures via `src/gate_d.py`.

ACKNOWLEDGMENT

(omitted in the double-anonymous version.)

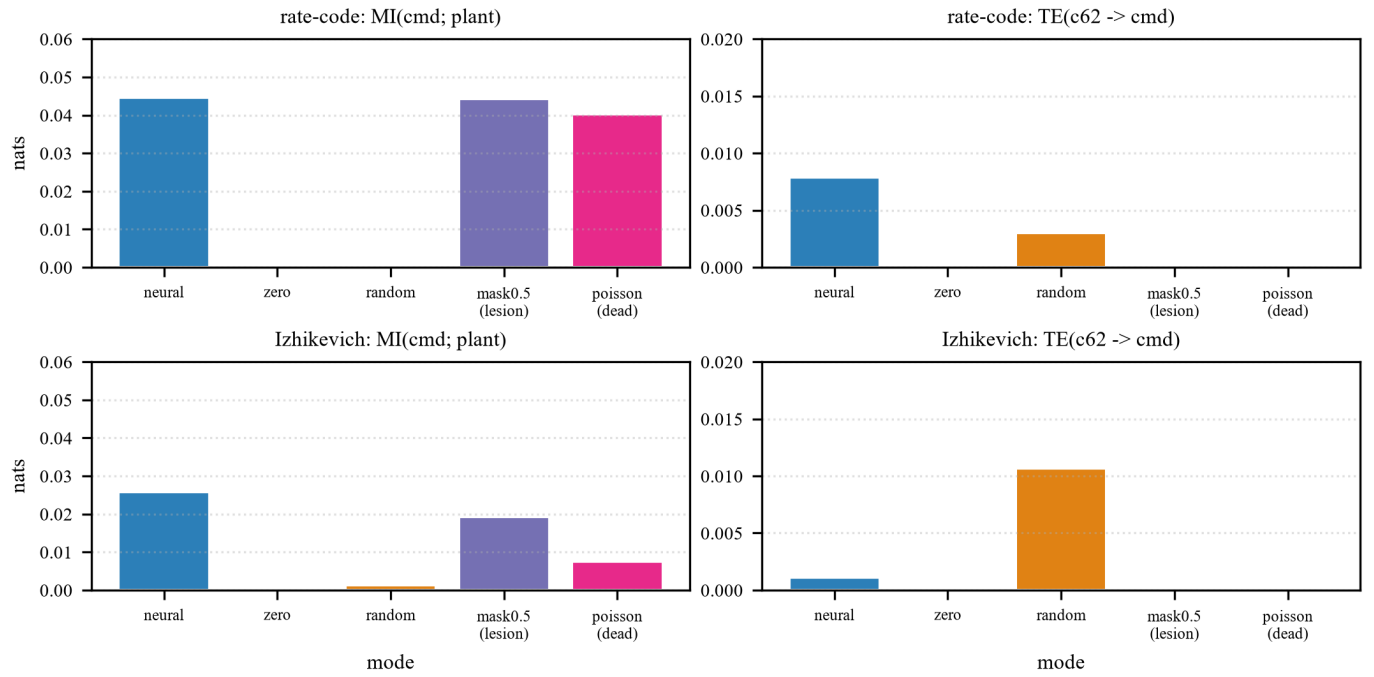


Fig. 5. Mutual information (left) and transfer entropy (right) for both substrates. TE along the task channel is the causal signature and is nonzero only for the intact loop.

REFERENCES

- [1] “Embodied Neurocomputation: A Framework for Interfacing Biological Neural Cultures with Scaled Task-Driven Validation,” preprint, arXiv:2605.13315, 2026.
- [2] B. J. Kagan *et al.*, “In vitro neurons learn and exhibit sentence when embodied in a simulated game-world,” *Neuron*, vol. 110, no. 22, pp. 3952–3969, 2022.
- [3] F. Balci, S. Ben Hamed, T. Boraud, S. Bouret, T. Brochier, C. Brun, *et al.*, “A response to claims of emergent intelligence and sentence in a dish,” *Neuron*, vol. 111, no. 5, pp. 604–605, 2023.
- [4] B. J. Kagan *et al.*, “Scientific communication and the semantics of sentence,” *Neuron*, vol. 111, no. 5, 2023. [letter]
- [5] H. Cai *et al.*, “Brain organoid reservoir computing for artificial intelligence,” *Nature Electronics*, vol. 6, pp. 1032–1039, 2023.
- [6] W. Maass, “Networks of spiking neurons: the third generation of neural network models,” *Neural Networks*, vol. 10, no. 9, pp. 1659–1671, 1997.
- [7] E. M. Izhikevich, “Simple model of spiking neurons,” *IEEE Trans. Neural Networks*, vol. 14, no. 6, pp. 1569–1572, 2003.
- [8] E. M. Izhikevich, “Which model to use for cortical spiking neurons?” *IEEE Trans. Neural Networks*, vol. 15, no. 5, pp. 1063–1070, 2004.
- [9] W. Maass, T. Natschlaeger, and H. Markram, “Real-time computing without stable states: a new framework for neural computation based on perturbations,” *Neural Computation*, vol. 14, no. 11, pp. 2531–2560, 2002.
- [10] C. Eliasmith and C. H. Anderson, *Neural Engineering: Computation, Representation, and Dynamics in Neurobiological Systems*. MIT Press, 2003.
- [11] T. Bekolay *et al.*, “Nengo: a Python tool for building large-scale functional brain models,” *Frontiers in Neuroinformatics*, vol. 7, art. 48, 2014.
- [12] T. Schreiber, “Measuring information transfer,” *Phys. Rev. Lett.*, vol. 85, no. 2, pp. 461–464, 2000.
- [13] A. Kraskov, H. Stögbauer, and P. Grassberger, “Estimating mutual information,” *Phys. Rev. E*, vol. 69, 066138, 2004.
- [14] S. Panzeri, R. Senatore, M. A. Montemurro, and R. S. Petersen, “Correcting for the sampling bias in spike-train information measures,” *J. Neurophysiology*, vol. 98, pp. 1064–1072, 2007.
- [15] B. L. Walker and K. A. Newhall, “Inferring information flow in spike-train data sets using a trial-shuffle method,” *PLoS ONE*, vol. 13, no. 11, e0206977, 2018.
- [16] J. Theiler, S. Eubank, A. Longtin, B. Galdrikian, and J. D. Farmer, “Testing for nonlinearity in time series: the method of surrogate data,” *Physica D*, vol. 58, pp. 77–94, 1992.
- [17] M. D. Ernst, “Permutation methods: a basis for exact inference,” *Statistical Science*, vol. 19, no. 4, pp. 676–685, 2004.
- [18] S. H. Hurlbert, “Pseudoreplication and the design of ecological field experiments,” *Ecological Monographs*, vol. 54, no. 2, pp. 187–211, 1984.
- [19] A. P. Georgopoulos, A. B. Schwartz, and R. E. Kettner, “Neuronal population coding of movement direction,” *Science*, vol. 233, pp. 1416–1419, 1986.
- [20] R. Q. Quiroga and S. Panzeri, “Extracting information from neuronal populations: information theory and decoding approaches,” *Nature Reviews Neuroscience*, vol. 10, pp. 173–185, 2009.
- [21] L. Paninski *et al.*, “Superlinear population encoding of dynamic hand trajectory in primary motor cortex,” *J. Neuroscience*, vol. 24, no. 39, pp. 8551–8561, 2004.
- [22] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: a physics engine for model-based control,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.
- [23] Unitree Robotics, *unitree_rl_gym: deployment code and pre-trained policies*, 2024. [Online]. Available: https://github.com/unitreerobotics/unitree_rl_gym